NAYOUNG PARK

Ph.D. Candidate | Computational Genomics & Bioinformatics

Konkuk University, Seoul, South Korea · p3159@konkuk.ac.kr · github.com/nayoung9
ORCID: 0000-0003-2722-0639 · linkedin.com/in/nayoung-park-75ba1b137

RESEARCH INTERESTS

Computational methods for genome interpretation, including cross-assembly annotation transfer, structural variation, read alignment quality, and population genomics; transcriptomic and single-cell analysis of disease.

EDUCATION

Ph.D. Candidate, Biomedical Science & Engineering · Konkuk University · Expected Feb 2027

M.S., Biomedical Science & Engineering (2020) · B.S., Stem Cell & Regenerative Biotechnology (2018)

SELECTED PUBLICATIONS

2026 Park N, Lee H, Kim J. scFlowReport: A Reproducible Workflow for Comparative Downstream Biological Analysis of Single-Cell RNA-seq Data. Biomolecules 16(9):1288.

2026 Park N, Lee J, et al. Type 17 T-cell subset divergence in hidradenitis suppurativa versus psoriasis. Frontiers in Immunology 17:1835460.

2026 Lee H, Park N, Park I, et al. Genome-wide characterization of AAV6 vector integration in human CD34+ cells. International Journal of Stem Cells.

2024 Park N, Kim H, Oh J, Kim J, et al. PAPipe: a pipeline for comprehensive population genetic analysis. Molecular Biology and Evolution 41(3).

2020 Lee J*, Park N*, Lee D, Kim J. Evolutionary and functional analysis of Korean native pig using SNPs. Molecules and Cells 43(8).

Selected publications; full publication list available on the website.

SELECTED ONGOING RESEARCH

synLiftOver Synteny-aware annotation transfer across genome assemblies and species.

PCNVBrowser Population-scale CNV analysis and interactive visualization across 26 human populations.

ROQ Machine-learning-based read overlap quality prediction for assessing alignment reliability.

SELECTED PRESENTATIONS & RECOGNITION

ASHG 2026 · Reviewer's Choice; Lightning Talk and Poster — synLiftOver | Montréal, Canada

RECOMB 2025 · Best Poster Award — MetaCNVNet | Seoul, South Korea

TECHNICAL SKILLS

Python, R, Bash, C++, SQL · scikit-learn, PyTorch, XGBoost · GATK, BCFtools, PLINK, Seurat, Scanpy · Snakemake, Nextflow, Docker, Git